Ali Suvizi

Ph.D. Researcher | Computer Architecture | FPGA/SoC | Hardware Accelerators
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RESEARCH PROFILE

Ph.D. researcher focused on computer architecture, reliable FPGA/SoC systems, chiplets, and heterogeneous accelerators. Experience leading FPGA/firmware prototype integration and developing C/C++ system-level simulators; first-author papers at ACM ICS and GLSVLSI, with cross-institution research collaboration.

EDUCATION

George Washington University, Washington, DC Aug 2023 – Dec 2027 (expected)
Ph.D. in Computer Engineering | Advisor: Prof. Guru Prasadh Venkataramani
Research Interest: High-Performance Computing, Computer Architecture, Machine Learning, Digital Design, LLMs

Amirkabir University of Technology, Tehran, Iran 2019 – 2022
M.Sc. in Computer Architecture

TECHNICAL SKILLS

Programming & systems: C, C++, Python, Bash; Linux, Docker; embedded ARM firmware, RISC-V.

Hardware & EDA: VHDL, Verilog, Vivado, Vitis HLS, ModelSim; Zynq-7000, AXI4/AXI4-Stream, DMA, partial and reconfiguration.

Architecture & simulation: Chiplets, Network-on-Chip (NoC), gem5, SST/BYOD, accelerator interfaces, performance and power profiling.

AI & accelerators: PyTorch, TensorFlow, CNNs, GRU prediction, Processing-in-Memory (PIM), digital–photonic matrix–vector acceleration.

RESEARCH EXPERIENCE

Graduate Research Assistant, George Washington University Aug 2023 – Present
Computer Architecture and Reliable Hardware Systems | Washington, DC

Swift-Healer | Firmware-Reconfigurable Self-Healing (ACM GLSVLSI 2026)

- Led FPGA/firmware prototype integration and evaluation on a **Xilinx Zynq-7000 SoC**, combining a pipelined 128×128 Sobel accelerator, AXI DMA, ARM Cortex-A9 firmware, and partial reconfiguration.
- Integrated GRU-based fault prediction (held-out **F1 = 0.87**, two control cycles ahead), multi-sensor detection, and clock/reset/reconfiguration recovery for clock- and voltage-glitch scenarios.
- Measured **0.03 ms/frame** accelerator latency at 200 MHz; recovery took **5 μs–5 ms** by action, restoring edge-detection F1 to approximately **0.96** in evaluated cases.

Auto-Healer | Chiplet-Based Fault-Tolerant Perception (ACM ICS 2025)

- Designed an FPGA-based self-healing architecture for CNN perception using redundancy, synchronization, and checkpoint-assisted recovery; reported mean repair times of **40 ns** (transient faults) and **120 ns** (permanent faults).

Hybrid Digital–Photonic Accelerators | SST/BYOD (2026)

- Extended the C++ SST/BYOD simulator with RISC-V accelerator commands, configuration, command/completion queues, and functional tests for neural-network offload.
- Profiled AlexNet FC6: **71.6 μs/job** simulated weight reads versus **2.09 ns/job** modeled optical matrix–vector execution, revealing the data-movement bottleneck.
- Reproduced **258,048** 8-bit ADC conversion codes in offline checks; static calibration reduced FC8 RMSE from **0.05561** to **0.03747** without clipping in the evaluated workload.

SELECTED RESEARCH PROJECTS

LPCR | Energy-Efficient Network-on-Chip Links (IEEE TCAD 2025)

- Developed information-theoretic code reordering for serial NoC links, reducing switching activity by up to **13.2%** and link energy by up to **9.9%** with less than 4.9% extra transmitted bits.

PIM and FPGA DNN Inference for Edge Servers (IEEE Cloud Summit 2024)

- Evaluated latency-sensitive LeNet-5 inference on CPU, simulated PIM, and FPGA platforms: reported **92**× CPU-to-PIM speedup at **0.36 W**, plus **0.02 ms** FPGA inference latency in the published configurations.

Information-Freshness-Aware Chiplet Scheduling (IEEE Cloud Summit 2026)

- Collaborated on FD-Hybrid, a topology-aware scheduler balancing Age of Information, inter-chiplet transfer latency, and real-time deadline misses; the paper received the **Best Paper Award**.

Multi-Vendor Chiplet Firmware Compatibility (Ongoing Research)

- Investigating isolated firmware runahead and hardware-interface invariants for transaction validity, causality, ownership, and post-update rejoin; developing a proposed runahead-twin approach to flag incompatibilities before activation.

Energy-Efficient LLM Fine-Tuning (IEEE OJCS 2025)

- Evaluated LoRA, BF16, activation checkpointing/offloading, and power capping for LLaMA-3.1, Phi-3, and Gemma on an NVIDIA RTX 4090; reported **over 40%** memory savings and **5.55%** average throughput loss at a 300 W versus 450 W power cap.

SELECTED PUBLICATIONS

- **A. Suvizi**, J. Iwu, K. Amberiadis, and G. Venkataramani. *Swift-Healer: Firmware-Reconfigurable Self-Healing for Remote Glitch-Injection on Autonomous Navigation Systems*. ACM GLSVLSI, 2026.
- E. Dogan, **A. Suvizi**, E. Erbayat, S. Subramaniam, and G. Venkataramani. *Exploring Information-Freshness v. Task-Latency Tradeoffs for Schedulers in Chiplet-based Hardware*. IEEE Cloud Summit, 2026.
- **A. Suvizi** and G. Venkataramani. *Auto-Healer: Self-Healing Hardware for Perception Stage Faults in Autonomous Driving Systems*. ACM ICS, 2025.
- M. Adelkhani, **A. Suvizi**, et al. *LPCR: Information-Theory-Based Low-Power Code Reordering for Serial Links in Network-on-Chip*. IEEE TCAD, 2025.
- K. Gogineni, **A. Suvizi**, and G. Venkataramani. *LLMs on a Budget: System-Level Approaches to Power-Efficient and Scalable Fine-Tuning*. IEEE Open Journal of the Computer Society, 2025.
- **A. Suvizi**, S. Subramaniam, T. Lan, and G. Venkataramani. *Exploring In-Memory Accelerators and FPGAs for Latency-Sensitive DNN Inference on Edge Servers*. IEEE Cloud Summit, 2024.

Submitted manuscript (IEEE TDSC): A. A. Malik, **A. Suvizi**, G. Venkataramani, and A. Aysu. *Emulated Integrity Replica: Enabling Self-Healing on FPGA SoCs via Hierarchical Twins*. Submitted to *IEEE Transactions on Dependable and Secure Computing*; arXiv preprint, 2026.

HONORS AND PROFESSIONAL SERVICE

Honors:

- **Best Paper Award, IEEE Cloud Summit (2026)**
- **IEEE DAC** Conference Travel Grant (2026).
- **GWU** University Fellowship (2023, 2024, 2025, 2026)
- **GEM5** Bootcamp Student Travel Grant (2024)
- Ranked 1st in the graduating class of the M.Sc. at Amirkabir University of Technology

Peer reviewer: IEEE Micro; IEEE Computer Architecture Letters; IEEE Computer; IEEE Transactions on Sustainable Computing; IEEE Access; IEEE CAL; Journal of Hardware and Systems Security.